

to operate with a supply of 1.8V. The ADC can store data for analysis, while the CPU wakes up at a later time. Unlike MSP430, QE128 does not have a DMA module and cannot move conversion data to RAM without CPU intervention.

QE128 does not have a wait instruction and therefore cannot differentiate between stop and wait modes. It has two bits that control the effect of a stop instruction to enable either stop or wait mode.

**ARM.** An ARM processor provides the facilities of dynamic power management (wait for interrupt mode) and static power management (leakage control). ARM-based MCUs take a different approach to atomic low-power transition.

ARM silicon vendors, such as Atmel, NXP and TI, integrate the standard ARM7 or ARM9 cores with the set of proprietary peripherals, such as interrupt and power-management controllers. The integration is loose in that the internal state of ARM core doesn't impact the peripherals. In particular, the core can disable interrupts internally but it doesn't effect the external power-management or interrupt controller, which provides another layer for disabling and enabling interrupts.

Because of this design, ARM-based MCUs allow transition to the LPM with interrupts disabled at the ARM-core level. Upon such a transition, the power-management controller stops the CPU clock for the ARM core, but any interrupt enabled at the interrupt controller level can start the CPU clock. As soon as the core starts running again, it can enable interrupts to achieve low interrupt latency.

**Cortex-M3.** Cortex-M3 is ARM's 32-bit RISC architecture and differs from the traditional ARM7 or ARM9 cores. It provides two low-power options: sleep and deep-sleep modes. In sleep and deep-sleep modes, the Cortex-M3 processor core and the memory subsystem are not clocked but the enabled peripherals are

clocked. Any interrupt returns the device to run mode from one of the sleep modes. BOD enable/disable and software control shutdown operation of individual peripherals is supported.

Cortex-M3 has a tighter integration of the MCU core with system power management and the nested vectored interrupt controller. It provides a special instruction, called WFI (wait for interrupt), to stop the CPU clock. This instruction can be used while interrupts are locked. The CPU stops executing the code after WFI instruction, but any interrupt enabled in the NVIC wakes the CPU up.

**CIP-51 MCU.** The basic CIP-51

power management unit configuration registers activates these modes. But, this must happen with interrupts enabled. Otherwise, CIP-51 locks-up. This means that it is impossible to transit to any of these modes atomically.

Any enabled interrupt can pre-empt idle processing after the interrupts are enabled but before the idle mode is entered. Therefore every interrupt invalidates the idle-mode transition. Thus any interrupt that pre-empts the idle loop, should disable the idle mode, which, in turn, accomplishes the goal of an interrupt-safe transition to idle mode.

**M16C.** M16 provides two low-power

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architecture supports two low-power levels (idle and stop), but nowadays C8051F92x-93x devices support additional LPMs of suspend and sleep. The power management unit allows the device to enter and wake up from the available power modes.

In idle mode, all peripherals are fully functional and any interrupt can wake up the CPU. Stop mode is a legacy 8051 LPM where a reset is required to wake up. Suspend is similar to stop mode, but it has a very fast wake-up time. Ultra-low-power mode is sleep mode having flexible wake-up resources.

In battery-powered systems, the system should spend as much time as possible in sleep mode in order to preserve the battery life. Though idle and suspend modes provide a very fast wake-up time, power saving in these modes will not be as much as in sleep mode.

Each analogue peripheral can be disabled when not in use or placed in an LPM. Digital peripherals such as timers and serial buses draw some power whenever these are not in use. They draw no power in sleep mode.

Setting the power control and

er modes: wait and stop. In wait mode, the CPU operation is stopped but oscillators do not stop. All oscillators stop in stop mode. The CPU and all built-in peripheral functions stop. This mode is the most effective in reducing power consumption.

The M16C 16-bit processor from Renesas supports the low-power wait mode, which is entered using a special wait instruction. However, the M16C datasheet states specifically that interrupts must be enabled before executing the wait instruction. So clearly the M16C doesn't support atomic transition to the wait mode. Like the 8051, the only option for the M16C is to somehow disarm transition to the wait mode in ISRs, to prevent the background loop from entering wait mode just after an interrupt. Thus any interrupt that pre-empts the idle loop disables wait mode, which accomplishes the goal of an interrupt-safe transition to idle mode. In contrast to the 8051, however, the M16C accomplishes the low-power mode transition using a special instruction. ●

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